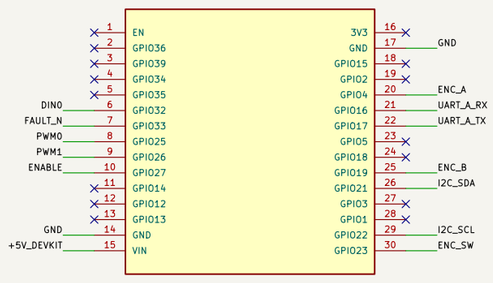
A1 — HOST AND COMMON INTERFACE / LOGIC 3.3 V

U1
ESP32 USB-C / 30P user module

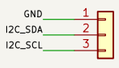


U1 is a removable DevKit, not a bare ESP32.
Common 30P pinout; row spacing25.40mm confirmed. Two HCTL PM254-1-15-Z-8.5 sockets.

J1
A-B POWER / XH-2



J6
A-B I2C / XH-3



J7
A-B CTRL / XH-6



J1 POWER: 1 GND / 2 +5V. J6 I2C: 1 GND / 2 SDA / 3 SCL.
J7 CTRL: 1 GND / 2 PWM0 / 3 ENABLE / 4 FAULT_N / 5 PWM1 / 6 DINO.
Straight-through cables, JST-XH 2.50 mm. NO 24 V, heater current or 3V3 rail.

#FLG01
PWR_FLAG



#FLG02
PWR_FLAG



#FLG03
PWR_FLAG



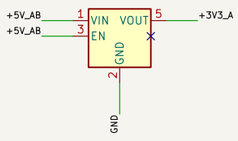
A2 — LOCAL POWER / LCD 3.3 V AS CURRENT PROTOTYPE

JP1
5V DevKit link / remove for service





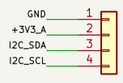
U2
AP2112K-3.3 peripherals







J2
LCD20x4 + backpack / 3V3



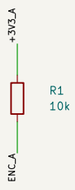
LCD header order is a proposed cable order.
Check actual backpack pin order, 3.3 V operation
and backlight current before PCB layout.

JP1 fitted in normal use. Do not parallel external 5 V and USB VBUS.
USB maintenance: B off, disconnect J1/J6/J7, remove JP1; no peripheral operation in this mode.
Powered serial debugging requires a verified VBUS-isolated USB connection. No automatic power mux.

A3 — USER INPUT / I2C PERIPHERALS

J3
Encoder / A B SW GND





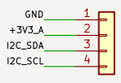




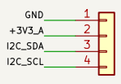




J4
TMP117 module / addr 0x48



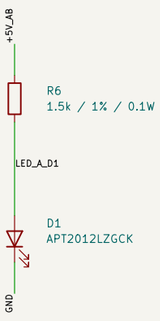
J5
BH1750 module / addr 0x23



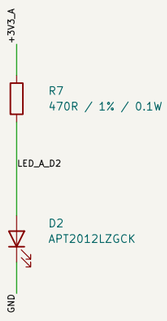
Only R4/R5 are the main I2C pull-ups. Remove or account for pull-ups on all breakout modules and LCD.
DevKit 3V3 is not joined to +3V3_A. U2 supplies peripherals only; check current and thermal budget.
I2C cable A-B: short internal harness. ADC on B = 0x49, TMP117 on A = 0x48.

DIAGNOSTICS — RAIL PRESENCE AND HEATER GATE DRIVE

D1 / 5V_AB



D2 / 3V3_A



Green LEDs show rail presence only, not regulated voltage accuracy or ESP32 execution.
HEAT (B/D8): 5 V driver output after PWM AND ENABLE; brightness follows duty, NOT measured heater current.
NTC on B/J4: 1 = +3V3_B, 2 = NTC_RAW; passive two-wire 15k B3920 probe, either wire orientation.

BENCH TEST PADS — BACK SIDE B.Cu — COPPER ONLY, DO NOT POPULATE

TP1
GND

TP2
+5V_AB

TP3
+5V_DEVKIT

TP4
+3V3_A

TP5
I2C_SDA

TP6
I2C_SCL

TP7
PWM0

TP8
ENABLE

TP9
FAULT_N

TP10
ENC_SW

UART EXTENSION — J10 — 3.3 V LOGIC, NOT RS232 VOLTAGE LEVELS

J10
A-B UART / 3V3 / GND TX_A RX_A



1 GND / 2 TX_A -> RX_B / 3 RX_A <- TX_B. Straight cable; no supply rail.

A: GPIO17 TX (U1.22), GPIO16 RX (U1.21); independent of USB UART0 and PWM.

Reserved extension; future B must implement its UART peripheral/transceiver. No hot-plug.

PROTOTYPE — ENGINEERING PREPARED — ASSEMBLER APPROVAL REQUIRED
Laboratory hardware platform
Sheet: /
File: pcb_a_display.kicad_sch
Title: PCB A — Display and ESP32 carrier
Size: User | Date: 2026-11-10 | **Rev: 0.4.3**
KiCad E.D.A. 10.0.6 | Id: 1/1